Experiment - 3 : Survey Report on Emerging Trends in Low-Code Application Development

Aim:

To study and analyse the emerging trends in low-code application development and understand their impact on modern software development.

Theory:

Low-code application development lets users create applications through visual interfaces, drag-and-drop components and minimal hand-written code. It reduces cost and development time and lets both professional and citizen developers build business applications such as a recruitment system. With growing demand for digital transformation, low-code platforms keep evolving. The major trends are:

- **Artificial Intelligence (AI) Integration:** AI assistants such as Zia can generate applications from a description, suggest fields and answer questions about data.
- **Citizen Development:** HR staff and other business users can build tools like a job application tracker with little programming knowledge.
- **Cloud-Native Development:** Applications are hosted in the cloud, giving easy deployment, scaling and remote access.
- **Business Process Automation:** Workflows, approvals and blueprints automate repetitive steps such as sending application confirmations.
- **Rich Visual Reporting:** Built-in Kanban, Calendar, Map, Timeline and Pivot views turn raw records into decisions.
- **Cross-Platform Applications:** The same app works on web, tablet and mobile without separate code.
- **API and Third-Party Integration:** Connections to email, payment, analytics and other services through APIs.
- **Security and Governance:** Role-based access, audit logs, environments and compliance features.
- **DevOps and Continuous Deployment:** Development, Stage and Production environments with version history.

Observation:

Trend	Benefit	Challenge
AI integration	Faster building and smarter insights	Accuracy depends on data; features vary by plan
Citizen development	Business users solve their own problems	Needs governance to avoid uncontrolled apps
Cloud-native	No server setup, easy scaling	Dependence on internet and vendor
Process automation	Less manual work	Complex logic still needs scripting
Visual reporting	Quick decisions from dashboards	Advanced analytics may need add-ons
API integration	Connects to existing systems	Requires API knowledge and security care

Trend	Benefit	Challenge
Security and governance	Controlled, compliant applications	Extra configuration effort

Procedure:

1 Study the concept of low-code application development.

2 Identify popular low-code platforms such as Zoho Creator, Microsoft Power Apps, OutSystems, Mendix and Salesforce Platform.

3 Collect information on recent trends from research papers, industry reports and official documentation.

4 Analyse trends such as AI integration, citizen development, cloud computing, automation, API integration and security.

5 Compare the benefits and challenges of each trend.

6 Record the observations and summarise the findings in the survey report.

Result:

The survey showed that low-code application development is evolving rapidly through AI-assisted development, cloud-native platforms, citizen development, process automation, rich visual reporting, API integration and stronger security. These trends improve development speed, reduce cost and help organisations build scalable applications for digital transformation.